The Oort cloud rules out the tidal screen at $C = 0$

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Abstract

The tidal screen of the companion paper, with the Lagrangian constant set to zero, puts new comets on the wrong side of the observed Oort peak. A comet that falls in from rest arrives with a Newtonian $1/a$ of about $1.23 \times 10^{-4}\,\mathrm{au}^{-1}$, so its original semimajor axis is at most about 8,100 au. The observed peak of new comets lies between 10^4 and 10^5 au. The push sits in a shell near 9 to 12 thousand au, where the net gravity on a test body points outward. Three evaluations of the same map agree at the percent level. A radial orbit integration at the same $C = 0$ lands the same way: starts of 3×10^4 and 9.5×10^4 au arrive at 4,877 and 6,270 au, and two starts inside the shell never reach the inner system. This paper records that null. It does not introduce a new constant.

1 What was frozen

The calculation was specified before the screened numbers were computed. The map uses radii from 0.05 to 300 kAU and twelve Gauss–Legendre nodes in the cosine of the angle from the external field. The external-field strengths are 1.9×10^{-10} and $2.32 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. The action-form screen is the one whose x^* was fixed by Cassini at 7.04×10^{11}. The constant C in the shell stress is zero, so the stress is proportional to $P(y)$, not to $P(y) - C$.

A comet falling from rest at infinity has an osculating Newtonian binding

$$\frac{1}{a} = \frac{2\Phi(0)}{GM_\odot},$$

with $\Phi(0)$ the line integral of the anomalous acceleration. Energy conservation fixes this number. The torque along the orbit is a separate statement and is not the kill.

2 The binding offset

At $g_e = 1.9 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ the offset is $+1.23 \times 10^{-4}\,\mathrm{au}^{-1}$. At 2.32×10^{-10} it is $+1.61 \times 10^{-4}\,\mathrm{au}^{-1}$. Every comet that the model places on a bound orbit from rest therefore has an original semimajor axis of at most about 8,100 au. The observed peak lies at 10^4–10^5 au, so the model leaves that peak empty.

evaluation	1.9×10^{-10}	2.32×10^{-10}
line integral, 12 directions	$+1.23 \times 10^{-4}$	$+1.61 \times 10^{-4}$
phantom-source potential at the Sun	$+1.227 \times 10^{-4}$	$+1.610 \times 10^{-4}$

The twelve directions agree with each other at the level of 0.1 percent. The phantom-source split is a divergence piece of -1.9×10^{-5} and a shell piece of $+1.42 \times 10^{-4}$. The shell raises the potential inside itself. A separate W-only radial integral, without the divergence piece, gives $1.386 \times 10^{-4}\,\mathrm{au}^{-1}$ at 1.9×10^{-10}, and an edge at 7,216 au. The two calculations are the same shell. They are not the same integral. The number that kills $C = 0$ here is the full map, $1.23 \times 10^{-4}\,\mathrm{au}^{-1}$.

3 Where the gravity points outward

The orientation-averaged anomalous acceleration, divided by the solar Newtonian acceleration, reaches +0.97 at 8.7 kAU, +1.27 at 10 kAU, and +0.99 at 11.5 kAU. Between about 8.7 and 11.5 kAU the net radial gravity on a test body points outward. A body placed there is not bound to the Sun in the usual sense.

Radial integrations at $C = 0$, with angular momentum aimed at a 5 au perihelion, give the same conclusion orbit by orbit. A start at 1.39×10^4 au turns at 9,683 au and does not reach the inner system. A start at 3.0×10^4 au reaches perihelion with osculating semimajor axis 4,877 au. A start at 9.49×10^4 au arrives at 6,270 au. Both arrivals sit below 10^4 au. The osculating $1/a$ matches the potential to $1.3 \times 10^{-6}\,\mathrm{au}^{-1}$ on the orbits that do reach perihelion.

4 Why a larger screen radius is not available

The offset scales as $1/x^*$. Cassini already sets the largest x^* the action form is allowed, 7.04×10^{11}. Moving x^* up to weaken the Oort push moves the Cassini quadrupole through its bound. There is no remaining x^* that keeps both. This paper does not retune a_0 or the interpolating function.

The torque proxy is consistent with the same shell and is not the kill. The jump semimajor axis is 3.67 kAU and 4.33 kAU at the two external fields, against 2.63 kAU with the screen removed and 26.9 kAU for Newton. Newton's jump falling between 15 and 40 kAU was the check that the integrator is the one that was pre-registered.

5 What remains open

A population synthesis of the comet catalogue, with planetary perturbations and the Galactic tide acting together over the age of the Solar System, has not been run at $C = 0$. The statement here is the energy integral, checked by direct radial orbits. Calling the observed peak empty is the comparison of that integral with the published range 10^4–10^5 au. The population synthesis is open.

The companion paper studies the same equation at the nonzero C that moves the edge into the observed window. That nonzero value is a different model. This paper does not adopt it. It records that $C = 0$ fails.